

3.2 Linear Momentum & Conservation

Question Paper

Course	CIEA Level Physics
Section	3. Dynamics
Topic	3.2 Linear Momentum & Conservation
Difficulty	Hard

Time allowed: 70

Score: /53

Percentage: /100

Question 1a

A stationary firework explodes into three different fragments that move in a horizontal plane. Fig. 1.1. shows the masses and velocities of the fragments A, B and C in terms of M and v .

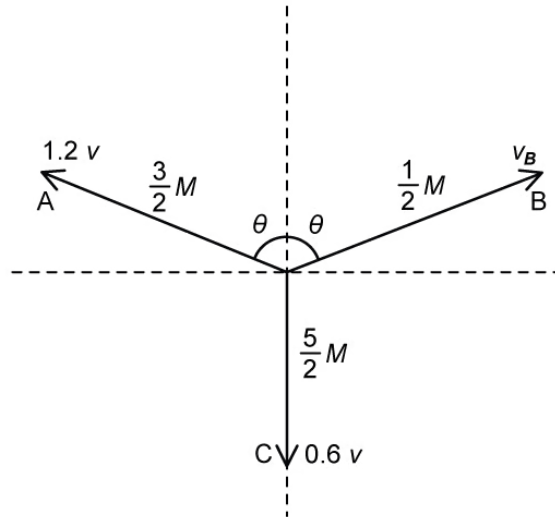


Fig. 1.1

Use the principle of conservation of momentum to determine the speed v_B of fragment B in terms of v .

[3 marks]

Question 1b

Use the principle of conservation of momentum to determine the angle θ .

[3 marks]

Question 1c

Another stationary firework explodes into four fragments which travel in different directions in a horizontal plane.

Fig. 1.2 shows the velocity and mass of each fragment.

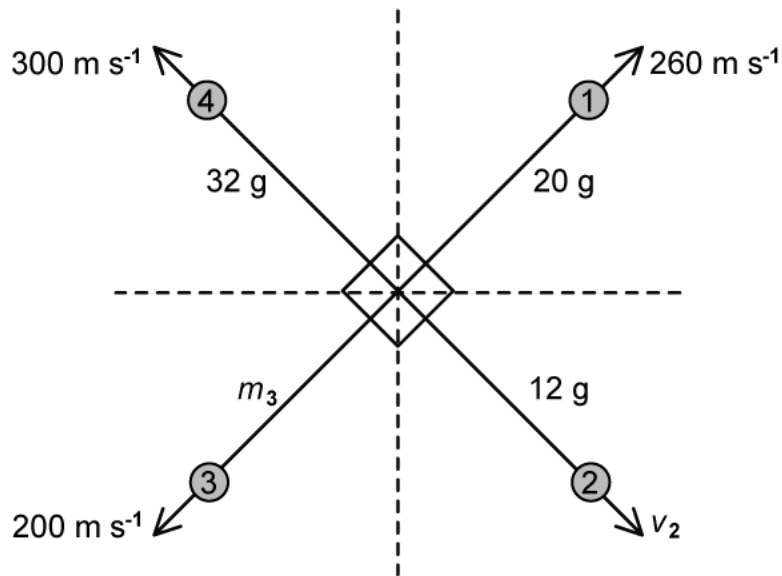


Fig. 1.2

Apply the principle of conservation of momentum to determine

(i)
the velocity v_2

[2]

(ii)
the mass m_3

[2]

[4 marks]

Question 1d

Both fireworks have the same initial mass and release the same amount of kinetic energy.

Calculate

(i)
the value of M

[1]

(ii)
the magnitude of v .

[3]

[4 marks]

Question 1e

Challenge

A firework similar to the one in (c) explodes into four fragments.

Fig. 1.3 shows the velocity and mass of each fragment.

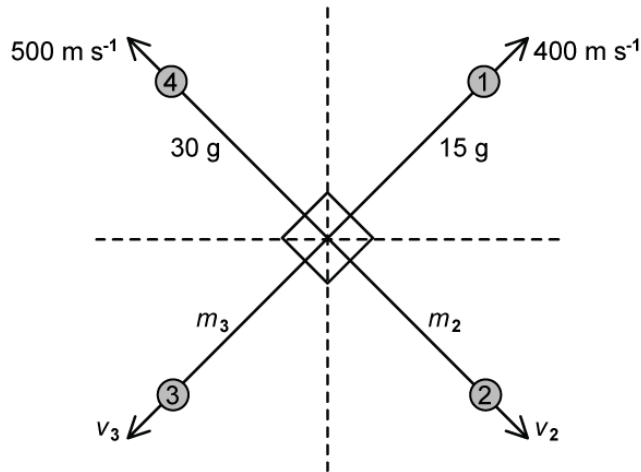


Fig. 1.3

(i)

By applying the principle of conservation of momentum, show that

$$\frac{m_1}{m_2} = \frac{v_3}{v_4} = \frac{3}{5}$$

$$\frac{m_3}{m_4} = \frac{v_1}{v_2} = \frac{2}{3}$$

[6]

(ii)

Hence, determine the values of m_2 , v_2 , m_3 and v_3 .

[2]

[8 marks]

Question 2a

A pellet of mass 2.7 g is fired into a stationary block of mass 540 g suspended from a rigid support as shown in Fig. 1.1.

The pellet becomes completely embedded in the block. The block can swing freely at the end of a light inextensible string of length 1.5 m measured from the pivot to the centre of the block.

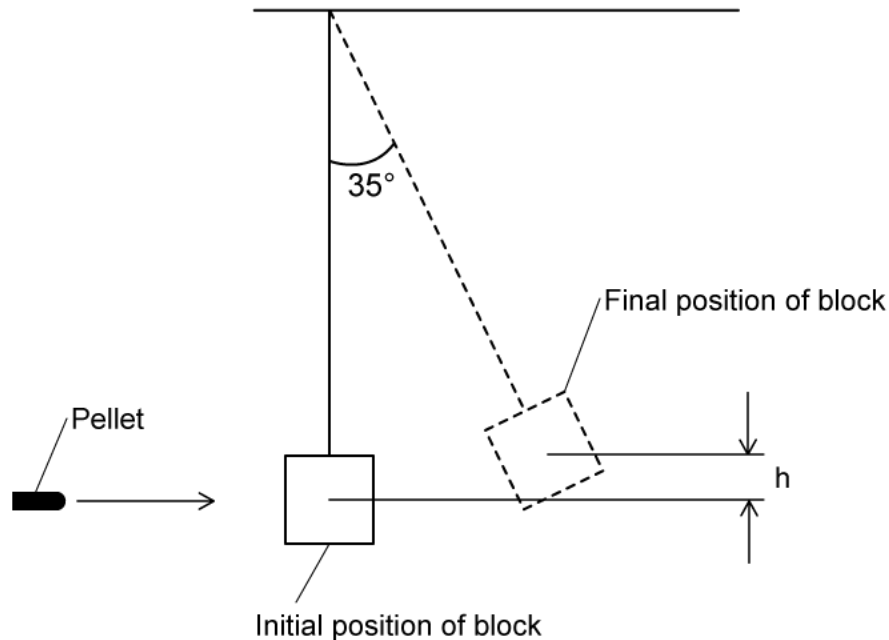


Fig. 1.1

The centre of mass of the block rises by a height h at an angle of 35° to the vertical.

Determine the velocity of the pellet and the block immediately after the pellet is embedded.

[4 marks]

Question 2b

Calculate the velocity of the pellet just before it strikes the block.

[3 marks]**Question 2c**

The block is replaced by a metal block of the same mass. The experiment is repeated with the wooden block and an identical pellet. The pellet rebounds after striking the block.

A student makes an assumption that the angle that the metal block makes with the vertical will be greater than 35° because the block doesn't have the additional mass of the pellet embedded within it.

Discuss the validity of the student's assumptions.

[6 marks]

Question 2d

A popular demonstration of the conservation of momentum and conservation of energy is Newton's cradle. It features several identical polished steel balls hung in a straight line in contact with each other, as shown in Fig. 1.2.

If one ball is pulled back and allowed to strike the line, one ball is released from the other end whilst the rest are stationary. If two are pulled out, two are released on the other end and so forth.

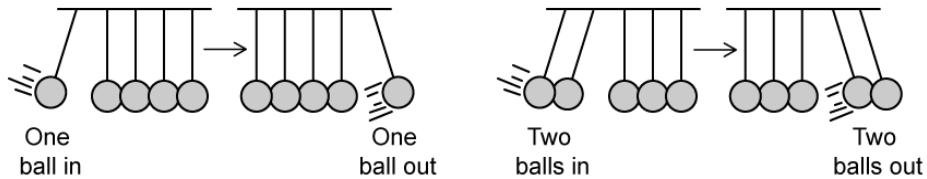


Fig. 1.2

Explain why swinging one ball from the left will not release two balls on the right.

Assume that the demonstration takes place in a vacuum.

[5 marks]

Question 3a

A body of mass M and moving with velocity V explodes into three equal fragments each of mass $\frac{M}{3}$, as shown in Fig. 1.1.

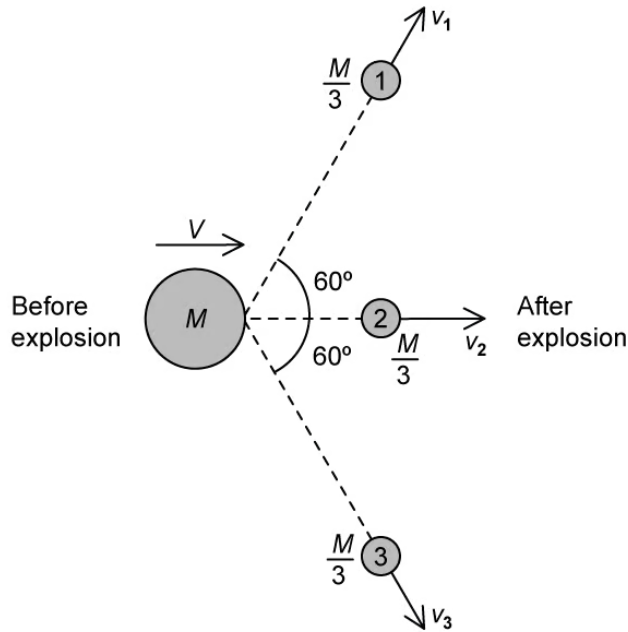


Fig. 1.1

Fragment 2 continues to move along the horizontal plane with velocity v_2 , while fragments 1 and 3 move off at an angle of 60° to the horizontal with velocities v_1 and v_3 respectively, as shown.

By applying the principle of conservation of momentum in the vertical plane, show that $v_1 = v_3$

[2 marks]

Question 3b

By applying conservation of momentum in the horizontal plane, show that $3V = (v_1 + v_2)$

[3 marks]

Question 3c

The ratio of the kinetic energy of fragment 1 to mass M is

$$\frac{\text{kinetic energy of fragment 1}}{\text{kinetic energy of mass } M} = \frac{4}{3}$$

Express velocities v_1 and v_2 in terms of the velocity V of mass M .

[4 marks]

Question 3d

(i)

Determine the total kinetic energy transferred to the fragments.

[3]

(ii)

Suggest why the total kinetic energy of the fragments is greater than the initial kinetic energy of the original mass.

[1]

[4 marks]

Model Answers: Hard

1a

(a) Determine the speed of fragment B:

Resolve the momenta horizontally:

Any **one** from:

- Horizontal momentum of A: $p_A = \frac{3}{2}M \times 1.2v \sin \theta$ [1 mark]
- Horizontal momentum of B: $p_B = \frac{1}{2}M \times v_B \sin \theta$ [1 mark]
- Horizontal momentum of C: $p_C = 0$ [1 mark]

Apply conservation of momentum in the horizontal direction:

- Conservation of momentum: total momentum before = total momentum after
Total momentum before = 0
Total momentum after = $p_A - p_B$
- Hence $p_A - p_B = 0 \Rightarrow p_A = p_B$
- Substitute: $\frac{3}{2}M \times 1.2v \sin \theta = \frac{1}{2}M \times v_B \sin \theta$ [1 mark]
- Simplify: $\frac{3}{2} \times 1.2v \times M \sin \theta = \frac{1}{2}v_B \times M \sin \theta \Rightarrow v_B = 2 \times \frac{3}{2} \times 1.2v$
- Speed of B: $v_B = 3.6v$ [1 mark]

[Total: 3 marks]

Remember that the momenta are vectors and as such have a specific sign for their direction.

$$+p_A \text{ and } -p_B$$

1b

(b) Determine the angle θ :

Resolve the momenta vertically:

Any **one** from:

- Vertical momentum of A: $p_A = \frac{3}{2}M \times 1.2v \cos \theta$ [1 mark]
- Vertical momentum of B: $p_B = \frac{1}{2}M \times 3.6v \cos \theta$ [1 mark]
- Vertical momentum of C: $p_C = \frac{5}{2}M \times 0.6v$ [1 mark]

Apply conservation of momentum in the vertical plane:

- Conservation of momentum: total momentum before = total momentum after
Total momentum before = 0
Total momentum after = $p_A + p_B - p_C$
- Therefore, $p_A + p_B - p_C = 0 \Rightarrow p_A + p_B = p_C$
- Substitute: $\frac{3}{2}M \times 1.2v \cos \theta + \frac{1}{2}M \times 3.6v \cos \theta = \frac{5}{2}M \times 0.6v$ [1 mark]
- Simplify: $\left(\frac{3}{2} \times 1.2 + \frac{1}{2} \times 3.6\right) \cos \theta = \frac{5}{2} \times 0.6$
- Rearrange: $\cos \theta = \frac{1.5}{3.6}$
- Angle: $\theta = 65.38 = 65^\circ$ [1 mark]

[Total: 3 marks]

1c

(c)

List the known quantities:

- Fragment (1): mass $m_1 = 20 \text{ g}$, velocity $v_1 = 260 \text{ m s}^{-1}$
- Fragment (2): mass $m_2 = 12 \text{ g}$, velocity $= v_2$
- Fragment (3): mass $= m_3$, velocity $v_3 = 200 \text{ m s}^{-1}$
- Fragment (4): mass $m_4 = 32 \text{ g}$, velocity $v_4 = 300 \text{ m s}^{-1}$

(i) Determine the velocity of fragment (2):

- Momentum of (4): $p_4 = m_4 v_4 = 32 \times 300$
- Momentum of (2): $p_2 = m_2 v_2 = 12 \times v_2$

Apply conservation of momentum:

- Conservation of momentum (taking p_2 as the positive direction):

$$0 = p_2 - p_4 \quad \Rightarrow \quad p_2 = p_4$$

- $12 \times v_2 = 32 \times 300$ [1 mark]

- $v_2 = \frac{32 \times 300}{12} = 800 \text{ m s}^{-1}$ [1 mark]

(ii) Determine the mass of fragment (3):

- Momentum of (1): $p_1 = m_1 v_1 = 20 \times 260$
- Momentum of (3): $p_3 = m_3 v_3 = m_3 \times 200$

Apply conservation of momentum:

- Conservation of momentum (taking p_1 as the positive direction):

$$0 = p_1 - p_3 \quad \Rightarrow \quad p_1 = p_3$$

- $20 \times 260 = m_3 \times 200$ [1 mark]

- $m_3 = \frac{20 \times 260}{200} = 26 \text{ g}$ [1 mark]

[Total: 4 marks]

This kind of explosion question can look daunting at first, but taking some time to understand what is going on reveals the problem can be simplified by isolating the fragments and focusing on each line of action one at a time

Since the fragments all move at 90° to one another, you can consider the directions independently, similar to taking 'horizontal' and 'vertical' components in other vector problems. Once you realise this, then you can see that the fragments on the same line have **equal momenta**

If you fancy a real maths challenge, have a go at part (e) of this question to see if you can truly get your head around how the momentum conservation laws can be applied to this sort of problem

1d

(d)

List the known quantities:

- Fragment (A): mass = $\frac{3}{2}M$, velocity = $1.2v$
- Fragment (B): mass = $\frac{1}{2}M$, velocity = $3.6v$
- Fragment (C): mass = $\frac{5}{2}M$, velocity = $0.6v$
- Fragment (1): mass = 20 g, velocity = 260 m s^{-1}
- Fragment (2): mass = 12 g, velocity = 800 m s^{-1}
- Fragment (3): mass = 26 g, velocity = 200 m s^{-1}
- Fragment (4): mass = 32 g, velocity = 360 m s^{-1}

(i) Calculate the value of M :

- Sum of the masses (from **(b)**) = $\frac{3}{2}M + \frac{1}{2}M + \frac{5}{2}M = \frac{9}{2}M$

- Sum of the masses (from **(c)**) = $20 + 12 + 26 + 32 = 90 \text{ g}$
- Therefore, $\frac{9}{2}M = 90 \text{ g} \Rightarrow M = \frac{90}{4.5} = 20 \text{ g}$ [1 mark]

(ii) Calculate the magnitude of v :

Total kinetic energy of fragments A, B and C:

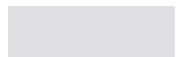
- $E_k = \frac{1}{2}Mv^2 \left[\left(\frac{3}{2} \times 1.2^2 \right) + \left(\frac{1}{2} \times 3.6^2 \right) + \left(\frac{5}{2} \times 0.6^2 \right) \right]$
- $E_k = 4.77Mv^2$ [1 mark]

Total kinetic energy of fragments 1, 2, 3 and 4:

- $E_k = \frac{1}{2}[(0.020 \times 260^2) + (0.012 \times 800^2) + (0.026 \times 200^2) + (0.032 \times 360^2)]$
- $E_k = 7110 \text{ J}$ [1 mark]

Determine the value of v :

$$\sqrt{\frac{7110}{4.77}}$$



Total kinetic energy of fragments 1, 2, 3 and 4:

$$\bullet E_k = \frac{1}{2}[(0.020 \times 260^2) + (0.012 \times 800^2) + (0.026 \times 200^2) + (0.032 \times 360^2)]$$

$$\bullet E_k = 7110 \text{ J [1 mark]}$$

Determine the value of v :

$$\bullet \text{ Total KE: } 4.77Mv^2 = 7110 \quad \Rightarrow \quad v = \sqrt{\frac{7110}{4.77 \times 0.020}}$$

$$\bullet v = 273 = 270 \text{ m s}^{-1} \text{ [1 mark]}$$

[Total: 4 marks]

The key is in the question:

- both fireworks have the **same initial mass**

This means the total mass of fragments A, B and C is equal to the total mass of fragments 1, 2, 3 and 4

- both fireworks release the **same amount of kinetic energy**

This means the total KE of fragments A, B and C is equal to the total KE of fragments 1, 2, 3 and 4

To get the total KE of fragments A, B and C, start by writing the kinetic energy of each in terms of M and v and then simplify:

$$E_k = \frac{1}{2} \left(\frac{3}{2} M \times (1.2v)^2 \right) + \frac{1}{2} \left(\frac{1}{2} M \times (3.6v)^2 \right) + \frac{1}{2} \left(\frac{5}{2} M \times (0.6v)^2 \right)$$

$$E_k = \frac{1}{2} M v^2 \left[\left(\frac{3}{2} \times (1.2)^2 \right) + \left(\frac{1}{2} \times (3.6)^2 \right) + \left(\frac{5}{2} \times (0.6)^2 \right) \right]$$

$$E_k = \frac{1}{2} M v^2 \times (9.54) = 4.77 M v^2$$

1e

(e)

(i) Apply conservation of momentum to Fig. 1.3:

- Momentum of fragment 1: $p_1 = m_1 v_1 = 0.015 \times 400 = 6 \text{ kg m s}^{-1}$
- Momentum of fragment 4: $p_4 = m_4 v_4 = 0.03 \times 500 = 15 \text{ kg m s}^{-1}$

From conservation of momentum:

- Fragments on the same line have equal momenta, therefore:

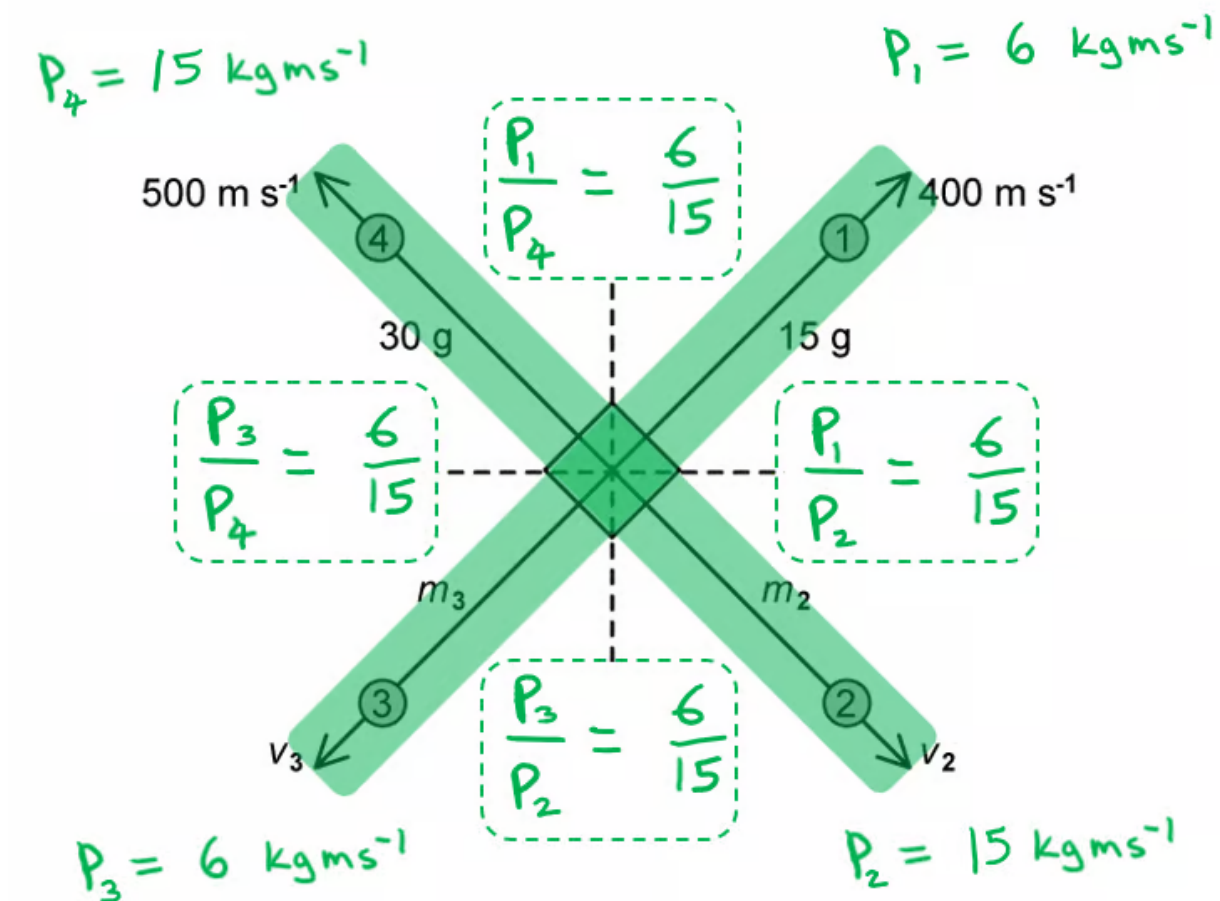
$$p_1 = p_3 = 6 \text{ kg m s}^{-1} \text{ [1 mark]}$$

$$p_2 = p_4 = 15 \text{ kg m s}^{-1} \text{ [1 mark]}$$

- Sum of $\uparrow$ momenta = sum of $\downarrow$ momenta: $p_1 + p_4 = p_2 + p_3$
- Sum of $\rightarrow$ momenta = sum of $\leftarrow$ momenta: $p_1 + p_2 = p_3 + p_4$
- So, $p_1 + p_4 = p_2 + p_3 = p_1 + p_2 = p_3 + p_4 = \text{constant}$
- Therefore, $\frac{p_1}{p_4} = \frac{p_2}{p_3} = \frac{p_1}{p_2} = \frac{p_3}{p_4} = \text{constant}$

- Hence, $\frac{m_1 v_1}{m_4 v_4} = \frac{m_3 v_3}{m_2 v_2} \text{ [Eq. 1] [1 mark]}$

- Hence, $\frac{m_1 v_1}{m_2 v_2} = \frac{m_3 v_3}{m_4 v_4} \text{ [Eq. 2] [1 mark]}$



List the known ratios so far:

- From Fig. 1.2: $\frac{m_1}{m_4} = \frac{15}{30} = \frac{1}{2}$
- From Fig. 1.2: $\frac{v_1}{v_4} = \frac{400}{500} = \frac{4}{5}$
- Since $p_1 = p_3$, then $\frac{m_1}{m_3} = \frac{v_3}{v_1}$
- Since $p_2 = p_4$, then $\frac{m_2}{m_4} = \frac{v_4}{v_2}$

- From **[Eq. 1]** $\frac{m_1}{m_4} \times \frac{v_1}{v_4} = \frac{m_3}{m_2} \times \frac{v_3}{v_2} = \frac{6}{15} = \frac{2}{5}$
- From **[Eq. 2]** $\frac{m_1}{m_2} \times \frac{v_1}{v_2} = \frac{m_3}{m_4} \times \frac{v_3}{v_4} = \frac{6}{15} = \frac{2}{5}$

List the known proportional relationships so far:

- $m_1 \propto v_3 \propto 1.5$ (simplified from 15 g)
- $v_1 \propto m_3 \propto 4$
- $m_4 \propto v_2 \propto 3$
- $v_4 \propto m_2 \propto 5$ (simplified from 500 m s⁻¹)

Use the proportional relationships to find the ratios in **[Eq. 1]**:

- $m_1 \propto v_3$ and $m_4 \propto v_2$ therefore $\frac{m_1}{m_4} = \frac{v_3}{v_2} = \frac{1.5}{3} = \frac{1}{2}$
- $v_1 \propto m_3$ and $m_2 \propto v_4$ therefore $\frac{m_3}{m_2} = \frac{v_1}{v_4} = \frac{4}{5}$

$$\frac{m_1 v_1}{m_4 v_4} = \frac{m_3 v_3}{m_2 v_2}$$

$$\frac{m_1}{m_4} \times \frac{v_1}{v_4} = \frac{m_3}{m_2} \times \frac{v_3}{v_2} = \frac{2}{5}$$

$m_4 = 30 \propto 3$
 $v_4 = 500 \propto 5$

$m_1 = 15 \propto 1.5$
 $v_1 = 400 \propto 4$

$\frac{m_1}{m_4} = \frac{15}{30} = \frac{1}{2}$
 $\frac{v_1}{v_4} = \frac{400}{500} = \frac{4}{5}$

$\frac{m_1}{m_3} = \frac{v_3}{v_1}$
 $\therefore m_1 \propto v_3$
 $m_3 \propto v_1$

$v_1 \propto m_3 \propto 4$

$\left(\frac{1}{2}\right) \times \left(\frac{4}{5}\right) = \frac{2}{5}$

$m_1 \propto v_3 \propto 1.5$

$\frac{m_3}{m_2} = \frac{v_1}{v_4}$
 $\therefore \frac{m_3}{m_2} = \frac{4}{5}$

$\frac{m_1}{m_4} = \frac{v_3}{v_2} = \frac{1}{2}$
 $\frac{v_1}{v_4} = \frac{m_3}{m_2} = \frac{4}{5}$

$\frac{v_3}{v_2} = \frac{m_1}{m_4}$
 $\therefore \frac{v_3}{v_2} = \frac{1}{2}$

Use the proportional relationships to find the ratios in **[Eq. 2]**:

- $v_4 \propto m_2 \propto 5$ (simplified from 500 m s^{-1})

Use the proportional relationships to find the ratios in **[Eq. 1]**:

- $m_1 \propto v_3$ and $m_4 \propto v_2$ therefore $\frac{m_1}{m_4} = \frac{v_3}{v_2} = \frac{1.5}{3} = \frac{1}{2}$
- $v_1 \propto m_3$ and $m_2 \propto v_4$ therefore $\frac{m_3}{m_2} = \frac{v_1}{v_4} = \frac{4}{5}$

Diagram illustrating the conservation of linear momentum in a collision. Two objects, 1 and 2, collide. Object 1 has mass $m_1 = 15 \text{ g}$ and initial velocity $v_1 = 400 \text{ m s}^{-1}$. Object 2 has mass m_2 and initial velocity v_2 . After the collision, object 1 has velocity v_3 and object 2 has velocity v_4 . Object 4 has mass $m_4 = 30 \text{ g}$ and initial velocity $v_4 = 500 \text{ m s}^{-1}$. Object 3 has mass m_3 and initial velocity v_3 .

Equations derived from the diagram:

$$\frac{m_1 v_1}{m_4 v_4} = \frac{m_3 v_3}{m_2 v_2}$$

$$\frac{m_1}{m_4} \times \frac{v_1}{v_4} = \frac{m_3}{m_2} \times \frac{v_3}{v_2} = \frac{2}{5}$$

From the diagram, the following ratios are identified:

- $m_4 = 30 \propto 3$
- $v_4 = 500 \propto 5$
- $m_1 = 15 \propto 1.5$
- $v_1 = 400 \propto 4$

Using these ratios, the following relationships are derived:

$$\frac{m_1}{m_4} = \frac{15}{30} = \frac{1}{2}$$

$$\frac{v_1}{v_4} = \frac{400}{500} = \frac{4}{5}$$

$$\frac{m_1}{m_3} = \frac{v_3}{v_1}$$

$$\therefore m_1 \propto v_3$$

$$\frac{m_3}{m_2} = \frac{v_1}{v_4}$$

$$\therefore m_3 \propto v_1$$

Combining the ratios:

$$\left(\frac{1}{2}\right) \times \left(\frac{4}{5}\right) = \frac{2}{5}$$

$$\frac{m_1}{m_4} = \frac{v_3}{v_2} = \frac{1}{2}$$

$$\frac{v_1}{v_4} = \frac{m_3}{m_2} = \frac{4}{5}$$

Final relationships:

- $v_1 \propto m_3 \propto 4$
- $m_1 \propto v_3 \propto 1.5$
- $\frac{v_3}{v_2} = \frac{m_1}{m_4}$
- $\therefore \frac{v_3}{v_2} = \frac{1}{2}$

Use the proportional relationships to find the ratios in **[Eq. 2]**:

- $m_1 \propto v_3$ and $m_2 \propto v_4$ and from the question $\frac{m_1}{m_2} = \frac{v_3}{v_4} = \frac{3}{5}$: therefore

$$\frac{m_1}{m_2} = \frac{v_3}{v_4} = \frac{1.5}{5} \times 2 = \frac{3}{5} \text{ [1 mark]}$$

- $m_3 \propto v_1$ and $m_4 \propto v_2$ and from the question $\frac{m_3}{m_4} = \frac{v_1}{v_2} = \frac{2}{3}$ therefore

$$\frac{m_3}{m_4} = \frac{v_1}{v_2} = \frac{4}{3} \div 2 = \frac{2}{3} \text{ [1 mark]}$$

Diagram illustrating the conservation of momentum in a collision between two objects, 1 and 2, and the resulting fragments 3 and 4.

Object 1 (Mass m_1): Initial velocity $v_1 = 400 \text{ m s}^{-1}$ (up-right). Fragment 1 (Mass m_1) moves up-right at 400 m s^{-1} . Fragment 2 (Mass m_2) moves down-right at v_2 .

Object 2 (Mass m_2): Initial velocity v_2 (down-right). Fragment 3 (Mass m_3) moves down-left at v_3 . Fragment 4 (Mass m_4) moves up-left at 500 m s^{-1} .

Conservation of Momentum Equations:

- Horizontal direction: $m_1 v_1 = m_1 v_1 + m_2 v_2$ (Note: This equation is not explicitly written in the diagram, but the horizontal components of momentum are conserved.)
- Vertical direction: $m_1 v_1 = m_1 v_1 + m_2 v_2$ (Note: This equation is not explicitly written in the diagram, but the vertical components of momentum are conserved.)
- From the question: $\frac{m_3}{m_4} = \frac{v_1}{v_2} = \frac{2}{3}$
- From the diagram: $\frac{m_3}{m_4} = \frac{v_3}{v_4} = \frac{1.5}{5} \times 2 = \frac{3}{5}$
- Therefore: $\frac{m_3}{m_4} = \frac{v_1}{v_2} = \frac{2}{3}$
- From the diagram: $\frac{m_3}{m_4} = \frac{v_3}{v_4} = \frac{1.5}{5} \times 2 = \frac{3}{5}$
- From the diagram: $\frac{m_3}{m_4} = \frac{v_3}{v_4} = \frac{1.5}{5} \times 2 = \frac{3}{5}$
- From the diagram: $\frac{m_3}{m_4} = \frac{v_3}{v_4} = \frac{1.5}{5} \times 2 = \frac{3}{5}$

Final Results:

- $m_1 = 15 \text{ g}$
- $v_1 = 400 \text{ m s}^{-1}$
- $m_2 = 25 \text{ g}$
- $v_2 = 300 \text{ m s}^{-1}$
- $m_3 = 10 \text{ g}$
- $v_3 = 200 \text{ m s}^{-1}$
- $m_4 = 15 \text{ g}$
- $v_4 = 500 \text{ m s}^{-1}$

(ii) Hence, the missing values are:

- $\frac{m_1}{m_2} = \frac{3}{5} \Rightarrow m_2 = \frac{5}{3} \times 0.015 = 0.025 \text{ (25 g)}$

AND

$$\bullet \quad \frac{v_3}{v_4} = \frac{3}{5} \quad \Rightarrow \quad v_3 = \frac{3}{5} \times 500 = 300 \text{ m s}^{-1} \text{ [1 mark]}$$

$$\bullet \quad \frac{m_3}{m_4} = \frac{2}{3} \quad \Rightarrow \quad m_3 = \frac{2}{3} \times 0.030 = 0.020 \text{ (20 g)}$$

AND

$$\bullet \quad \frac{v_1}{v_2} = \frac{2}{3} \quad \Rightarrow \quad v_2 = \frac{3}{2} \times 400 = 600 \text{ m s}^{-1} \text{ [1 mark]}$$

[Total: 8 marks]

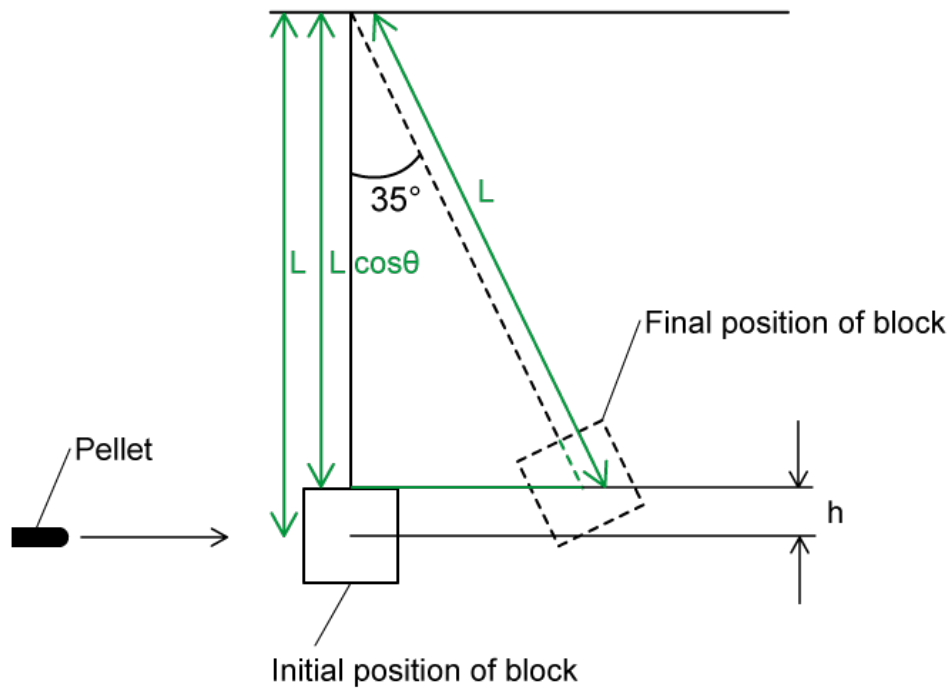
2a

(a) Determine the velocity of the pellet and the block immediately after the pellet is embedded:

List the known quantities:

- Mass of pellet = 2.7 g
- Mass of block = 540 g
- Combined mass, $M = 540 + 2.7 = 542.7 \text{ g}$
- Angle between string and vertical, $\theta = 35^\circ$
- Length of string, $L = 1.5 \text{ m}$

Calculate the change in height, h :



- From the diagram: $L = L \cos \theta + h \Rightarrow h = L - L \cos \theta$
- $h = 1.5 - 1.5 \cos 35^\circ = 0.271 \text{ m}$ [1 mark]

Calculate the initial speed of the block and the pellet:

- From conservation of energy: Initial KE = final GPE
- Initial KE = $\frac{1}{2}Mu^2$ **AND** final GPE = Mgh [1 mark]
- Equate and rearrange: $\frac{1}{2}Mu^2 = Mgh \Rightarrow u = \sqrt{2gh}$
- Substitute: $u = \sqrt{2 \times 9.81 \times 0.271}$ [1 mark]
- Initial speed of pellet and block: $u = 2.306 = 2.3 \text{ m s}^{-1}$ [1 mark]

[Total: 4 marks]

This type of question is known as a 'ballistic pendulum' and the marks are given for the method, as well as the final answer, so make sure you set out your working clearly!

The 'initial kinetic energy' is treated as when the pellet first strikes and completely embeds itself in the block. Therefore, the mass is always the **sum** of the mass of the

pellet and block.

You can see that when you equate KE and GPE algebraically, the mass term cancels, so if you're confident doing this, it will definitely save you some time when doing the calculation step!

2b

(b) Calculate the velocity of the pellet just before it strikes the block:

List the known quantities:

- Mass of the pellet, $m_p = 2.7 \text{ g} = 0.0027 \text{ kg}$
- Mass of the block, $m_b = 540 \text{ g} = 0.540 \text{ kg}$
- Combined mass, $M = 542.7 \text{ g} = 0.5427 \text{ kg}$
- Initial velocity of pellet and block, $u = 2.306 \text{ m s}^{-1}$ (from **(a)**)

Apply the principle of conservation of momentum:

- Total momentum lost by pellet = total momentum gained by pellet and block
- Total momentum lost by pellet = $m_p v_p$
- Total momentum gained by pellet and block = $(m_p + m_b)u = Mu$
- $m_p v_p = (m_p + m_b)u \Rightarrow m_p v_p = Mu$ [1 mark]

- Mass of the block, $m_b = 540 \text{ g} = 0.540 \text{ kg}$
- Combined mass, $M = 542.7 \text{ g} = 0.5427 \text{ kg}$
- Initial velocity of pellet and block, $u = 2.306 \text{ m s}^{-1}$ (from **(a)**)

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- Total momentum lost by pellet = total momentum gained by pellet and block
- Total momentum lost by pellet = $m_p v_p$
- Total momentum gained by pellet and block = $(m_p + m_b)u = Mu$
- $m_p v_p = (m_p + m_b)u \Rightarrow m_p v_p = Mu$ [1 mark]
- $v_p = \frac{Mu}{m_p} = \frac{0.5427 \times 2.306}{0.0027}$ [1 mark]
- $v_p = 463.506 = 464 \text{ m s}^{-1}$ [1 mark]

[Total: 3 marks]

2c

(c) For the assumption:

The angle that the metal block makes with the vertical will be greater than 35° because the block doesn't have the additional mass of the pellet embedded within it

State whether the student's reasoning is correct or incorrect:

- The student is correct that the angle will increase; [1 mark]
- (However) the student's explanation / reasoning is incorrect; [1 mark]

Some valid comments about the student's assumption are:

Any **two** pairs from:

- The change in momentum of the metal block will be greater; [1 mark]
- (Because) a rebounding pellet causes a greater change in momentum; [1 mark]

- The speed / kinetic energy of the metal block will be greater (after the pellet has rebounded); [1 mark]
- (Because) the momentum / mass / speed of the pellet is still the same; [1 mark]
- (From conservation of energy) all the initial KE is transferred to the GPE of the block; [1 mark]
- (Hence) the height reached by the metal block will be greater (than the height reached by the original block); [1 mark]

[Total: 6 marks]

2d

(d) Swinging one ball from the left will not release two balls on the right because:

- This would violate the conservation of energy / energy will not be conserved;
OR
- It is an inelastic collision but should be elastic (in a vacuum); [1 mark]

Apply conservation of momentum:

- Conservation of momentum: total momentum before = total momentum after
- Total momentum before = mu
- Total momentum after = $2mv$
- Conservation of momentum: $mu = 2mv \Rightarrow v = \frac{u}{2}$ [1 mark]
- Therefore, conservation of momentum is not violated if the final speed of the two balls is halved; [1 mark]

Apply conservation of energy:

- In a vacuum, there is no energy lost to the surroundings
- Hence, elastic collision: total kinetic energy before = total kinetic energy after
- Total kinetic energy before = $\frac{1}{2}mu^2$
- Total kinetic energy after = $\frac{1}{2}(2m)\left(\frac{u}{2}\right)^2 = \frac{1}{4}mu^2$ [1 mark]

- Therefore, conservation of energy is violated as half the kinetic energy is lost; [1 mark]

[Total: 5 marks]

3a

(a) Show that $v_1 = v_3$ by considering momentum in the vertical plane:

- Vertical plane: total momentum before = 0
- Vertical plane: total momentum after = $\left(\frac{M}{3}\right)v_1 \sin 60^\circ - \left(\frac{M}{3}\right)v_3 \sin 60^\circ$ [1 mark]

Apply conservation of momentum:

- Conservation of momentum: total momentum before = total momentum after
- $0 = \left(\frac{M}{3}\right)v_1 \sin 60^\circ - \left(\frac{M}{3}\right)v_3 \sin 60^\circ$
- Simplify: $\left(\frac{M}{3}\right)v_1 \sin 60^\circ = \left(\frac{M}{3}\right)v_3 \sin 60^\circ \Rightarrow v_1 = v_3$ [1 mark]

[Total: 2 marks]

3b

(b) Show that $3V = (v_1 + v_2)$ by considering momentum in the horizontal plane:

- Horizontal plane: total momentum before = MV
- Horizontal plane: total momentum after = $\left(\frac{M}{3}\right)v_1 \cos 60^\circ + \left(\frac{M}{3}\right)v_2 + \left(\frac{M}{3}\right)v_3 \cos 60^\circ$ [1 mark]

Apply conservation of momentum:

- Conservation of momentum: total momentum before = total momentum after

- $MV = \left(\frac{M}{3}\right)v_1 \cos 60^\circ + \left(\frac{M}{3}\right)v_2 + \left(\frac{M}{3}\right)v_3 \cos 60^\circ$
- Substitute $v_1 = v_3$ in: $MV = 2 \times \left(\frac{M}{3}\right)v_1 \cos 60^\circ + \left(\frac{M}{3}\right)v_2$ [1 mark]
- Simplify: $V = 2 \times \left(\frac{1}{3}\right)v_1 \cos 60^\circ + \left(\frac{1}{3}\right)v_2 \Rightarrow 3V = (v_1 + v_2)$ [1 mark]

[Total: 3 marks]

The full worked solution is shown below:

$$MV = \left(\frac{M}{3}\right)v_1 \cos 60^\circ + \left(\frac{M}{3}\right)v_2 + \left(\frac{M}{3}\right)v_3 \cos 60^\circ$$

$$V = 2 \times \left(\frac{1}{3}\right)v_1 \cos 60^\circ + \left(\frac{1}{3}\right)v_2$$

$$3V = 2v_1 \cos 60^\circ + v_2$$

$$\cos 60^\circ = \frac{1}{2}$$

$$3V = 2 \times \frac{1}{2}v_1 + v_2$$

$$3V = (v_1 + v_2)$$

3c

(c) Express velocities v_1 and v_2 in terms of V :

- Kinetic energy of mass M : $KE_i = \frac{1}{2}MV^2$
- Kinetic energy of fragment 1: $KE_1 = \frac{1}{2}\left(\frac{M}{3}\right)v_1^2$ [1 mark]

Substitute the expressions into the ratio:

- $\frac{KE_1}{KE_i} = \frac{4}{3} \Rightarrow KE_1 = \frac{4}{3} KE_i$
- $\frac{1}{2} \left(\frac{M}{3} \right) v_1^2 = \frac{4}{3} \left(\frac{1}{2} M V^2 \right)$ [1 mark]

Simplify to determine the velocity of fragment 1:

- $\frac{1}{3} v_1^2 = \frac{4}{3} V^2 \Rightarrow v_1^2 = 4 V^2$
- Velocity of fragment 1: $v_1 = 2V$ [1 mark]

Determine the velocity of fragment 2:

- $3V = v_1 + v_2 \Rightarrow v_2 = 3V - 2V$
- Velocity of fragment 2: $v_2 = V$ [1 mark]

[Total: 4 marks]

3d

(d)

- Velocity of fragment 1: $v_1 = 2V$ [1 mark]

Determine the velocity of fragment 2:

- $3V = v_1 + v_2 \Rightarrow v_2 = 3V - 2V$
- Velocity of fragment 2: $v_2 = V$ [1 mark]

[Total: 4 marks]

3d

(d)

(i) Determine the total kinetic energy transferred to the fragments:

- Kinetic energy of fragment 1: $KE_1 = \frac{4}{3}KE_i$
- Kinetic energy of fragment 3: $KE_3 = \frac{4}{3}KE_i$ (since $v_1 = v_3$)
- Where Kinetic energy of mass M : $KE_i = \frac{1}{2}MV^2$ (from part (c))

Determine the proportion of KE transferred to fragment 2:

- $\alpha KE_i = KE_2$ where α is the proportion of KE transferred to the fragment
- $\alpha \left(\frac{1}{2}MV^2 \right) = \frac{1}{2} \left(\frac{M}{3} \right) V^2$ (since $v_2 = V$)
- $\alpha \left(\frac{1}{2}MV^2 \right) = \frac{1}{3} \left(\frac{1}{2}MV^2 \right) \Rightarrow \alpha = \frac{1}{3}$
- Therefore, $KE_2 = \frac{1}{3}KE_i$ [1 mark]

Determine the total kinetic energy:

- Total initial KE = $KE_1 + KE_2 + KE_3$
- Total initial KE = $\frac{4}{3}KE_i + \frac{1}{3}KE_i + \frac{4}{3}KE_i$ [1 mark]
- Total initial KE = $3KE_i$ [1 mark]

(ii) The kinetic energy of the fragments is greater than the initial kinetic energy because:

- The total energy before the explosion is equal to the kinetic energy of mass M plus...

Any **one** from:

- Chemical energy (stored in mass M); [1 mark]
- Thermal energy (from the explosion); [1 mark]
- Internal energy (stored in mass M); [1 mark]
- Work done (from the explosion); [1 mark]

[Total: 4 marks]

